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Assignment 5

1. Recurrence

$T(n) = 5T\left(\frac{n}{7}\right) + 3n^2$ expands to

- Substitutions

Substitution 1 result: $T(n) = 5^2T\left(\frac{n}{49}\right) + 3n^2\left(\frac{5}{49}\right)^2 + 3n^2$

Substitution 2 result: $T(n) = 5^3T\left(\frac{n}{7^3}\right) + 3n^2\left(\frac{5}{49}\right)^2 + 3n^2\left(\frac{5}{49}\right) + 3n^2$

- Pattern:

After i substitutions it is:

$$T(n) = 5^i \times T\left(\frac{n}{7^i}\right) + 3n^2 \times \Sigma\left(\frac{5}{49}\right)^k, \text{ for } k = 0 \text{ to } i - 1$$

Base case is reached when $\log_7 n$

$$T(n) = 5^{(\log_7 n)} \times T(1) + 3n^2 \Sigma\left(\frac{5}{49}\right)^k$$

Since $\left(\frac{5}{49}\right) < 1$, $5^{(\log_7 n)}$ simplifies to $n^{(\log_7 5)}$. This grows slower than n^2 , which will rule.

So, $T(n) = O(n^2)$

- Master Theorem:
 - $a=5, b=7, d=2$
 - Since $a < b^d$, $T(n) = \theta(n^d)$ can be applied $\rightarrow T(n) = \theta(n^2)$

2. Master Theorem

a. $T(n) = 5T\left(\frac{n}{9}\right) + n^7$

- $a = 5, b = 9, d = 7$
- $b^d = 9^7$, and since $a < b^d$, $T(n) = \theta(n^d)$ can be applied
- ANSWER: $T(n) = \theta(n^7)$

b. $T(n) = 3T\left(\frac{n}{3}\right) + 7n$

- $a = 3, b = 3, d = 1$

- $b^d = 3$, and since $a = b^d$, $T(n) = \theta(n^d \log n)$ can be applied
- ANSWER: $T(n) = \theta(n \log n)$
- c. $T(n) = 15T\left(\frac{n}{4}\right) + 100$
 - $a = 15$, $b = 4$, $d = 0$
 - $b^d = 1$, and since $a > b^d$, $T(n) = \theta(n^{\log b a})$ can be applied
 - ANSWER: $T(n) = \theta(n^{\log_4 15})$
- d. $T(n) = 9T\left(\frac{n}{3}\right) + n^2$
 - $a = 9$, $b = 3$, $d = 2$
 - $b^d = 9$, and since $a = b^d$, $T(n) = \theta(n^d \log n)$ can be applied
 - ANSWER: $T(n) = \theta(n^2 \log n)$
- e. $T(n) = 6T\left(\frac{n}{5}\right) + n^3$
 - $a = 6$, $b = 5$, $d = 3$
 - $b^d = 125$, and since $a < b^d$, $T(n) = \theta(n^d)$ can be applied
 - ANSWER: $T(n) = \theta(n^3)$

3. Double Hashing.

BOY,RIS,GUA,ATL,SUC,CES,QUI,PUT,ANT,VAU,HUI,VAL,SAN,SAN,CAQ,VIC,CAL,LA_GUA, TOL,META,CAU,BOL,AMA,MAG,CHO,NSA,GUA,CAS,COR

- Pass 1: rightmost character is sorted into buckets

A	C	G	I	L	N	O	Q	R	S	T	U	Y
GUA, LA_GUA, A, META, AMA, NSA, GUA	SUC, VIC	MAG	QUI, HUI	ATL, VAL, CAL, TOL,BOL	SAN, SAN	CHO	CAQ	COR	RIS, CES, CAS	PUT, ANT	VAU, CAU	BOY

- Pass 2: A,E,H,I,M,N,O,S,T,U

A	E	H	I	M	N	O	S	T	U
MAG, VAL, CAL, SAN, SAN, CAQ, CAS,	CES	CHO	VIC, RIS	AMA	ANT	TOL, BOL, COR, BOY	NSA	META, ATL	GUA, LA_GUA, GUA,SUC,QUI,HUI, PUT

VAU, CAU									
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- Pass 3: A,B,C,E,G,H,M,N ,P,Q,R,S,T,V

A	B	C	E	G	H	M	N	P	Q	R	S	T	V
AMA,A NTATL	BOL, BOY	CAL, CAQ, CAS, CAU, CES, CHO, COR	ME TA	GUA,LA _GUA	H U I	M A G	N S A	P U T	Q U I	R IS	SA N, SA N, SU C	T O L	VAL, VAU, VIC

- Pass 4:

M	-
META	LA_GUA

- Pass 5:

A
LA_GUA

Pass 6:

L
LA_GUA

Final Sorted List: AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, HUI, LA_GUA, MAG, META, NSA, PUT, QUI, RIS, SAN, SUC, TOL, VAL, VAU, VIC

4. Double Hashing

[62, 62, 69, 70, 4, 103, 85, 27, 86, 40, 96, 77, 23]

- Key=62

1) slot 5 , 2) No collision, 3)

0	1	2	3	4	5	6
					62	

- Key=62

1) slot 5 , 2) 1 collision, so insert at slot 0, 3)

0	1	2	3	4	5	6
62					62	

- Key=69

1) Slot 5, 2) 2 collisions, so insert at slot 2, 3)

0	1	2	3	4	5	6
62		69			62	

- Key=70

1) Slot 1, 2) No collisions, 3)

0	1	2	3	4	5	6
62	70	69			62	

- Key=4

1) Slot 2, 2) 3 collisions, so insert at slot 3, 3)

0	1	2	3	4	5	6
62	70	69	4		62	

- Key=103

1), Slot 3 2), resize to M=14 , insert at slot 11 3)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	
	70		103		69				4		62		62	

- Key=85

1) Slot 7, 2) no collisions, 3)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	
	70		103		69				4		62		62	

- Key=27

1) Slot 5, 2) resize to m =28, rehash, 3) (5,69), (7,27), (9,4), (15,70), (17,103), (21,62), (25,85)

- Key=86

1) Slot 6, 2) no collisions, 3) (5,69), (6,86), (7,27), (9,4), (15,70), (17,103), (21,62), (25,85)

- Key=40

- 1) Slot 9, 2) 1 collision, so insert at slot 1, 3) (1, 40), (5,69), (6,86), (7,27), (9,4), (15,70), (17,103), (21,62), (25,85)
 - Key=96
- 1) Slot 0, 2) No collision, 3) (0,96), (1, 40), (5,69), (6,86), (7,27), (9,4), (15,70), (17,103), (21,62), (25,85)
 - Key=77
- 1) Slot 9, 2) 2 collisions so insert at slot 13, 3) (0,96), (1, 40), (5,69), (6,86), (7,27), (9,4), (13,77), (15,70), (17,103), (21,62), (25,85)
 - Key=23
- 1) 1) slot 0, 2) 1 collision, so insert at slot 24, 3) (0,96), (1, 40), (5,69), (6,86), (7,27), (9,4), (13,77), (15,70), (17,103), (21,62), (24,23), (25,85)

7. Algorithm Analysis

Question 5 – Radix Sort:

- The time complexity is $O(n*m)$ where n is the number of strings, and m is the maximum string length because every iteration is performed for every character in a string until all n strings are processed.
- The space complexity is $O(n)$ where n is the number of strings because as the input size increases or decreases, the algorithm will store all of that amount of strings, growing linearly.

Question 6 – Palindrome Triplets

- The time complexity is $O(n^3 \times k)$ where n is the number of words and k is the length of the strings once concatenated, because the three nested loops iterate through all possible triplets and then for each, the algorithm checks whether the strings are palindromes.
- The space complexity is $O(n^3)$ where n is the number of words, because as the input size changes in the worst case, when every combination results in a palindrome, the number of stored palindromes grows cubically with the input size.